

# Manufacturing Execution System

## Capability & Operations Handbook

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A reference, not a proposal. What the resin pouring station's production system does today, screen by screen — most of it optional, all of it built on one hourly log — with a working guide for each role and an honest account of what is still experimental.

# What's in this handbook

Sections 1 to 8 describe what the system does. Sections 9 to 11 are working guides for each role, written so a new starter can follow them on their first shift. Section 12 covers readability on the floor, and section 14 what exists but is not in service.

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## ABOUT THE SCREENSHOTS

Every screen shown is a live capture of the running system against a representative fortnight of activity. The operator and manager names in them are fictional. Most are in the dark house theme with their black floor lifted so the page can be printed without soaking the sheet; the two fill-weight figures are captured in **Paper White**, one of the light themes described in section 12, because lifting a dark capture washes out exactly the small labels those two are about.

# What the system is

Resin pouring has never had a production tracking system. Counts, downtime, changeovers and cartridge lots lived on paper and in people's heads, which means the plant has been running without a record it could query, an hourly pace it could see, or any way to answer a quality question after the fact.

This is that record. A web application opened in a browser — on an operator's phone at the pump, on a desk in the office, and on the floor display — backed by a PostgreSQL database. Operators log the hour as it happens, at the pump. That log is the whole input: the figures on the wall display, in the analytics charts and in the emailed shift handover are not separate reports, they are the same rows read different ways.

Nothing here asks a manager to enter anything. The system installs with work orders switched off — a plant that wants to dispatch runs to stations and track them against a target turns that on, and the screens for it appear; a plant that does not is a log, and the operator terminal never mentions runs at all. Every capability described in this handbook past section 3 works the same way: a setting, answering a question somebody asked first, not a step in getting the record.

**18**

screens, from the operator terminal to the IT console

**3**

roles, each seeing only what its job needs

**18**

database tables of production, quality and configuration

**2**

shifts, a plant setting rather than an assumption in the code

**34**

interface themes, six of them light, for readability in different conditions

**880**

automated checks run before a release, including a browser driven through a shift

## What it records

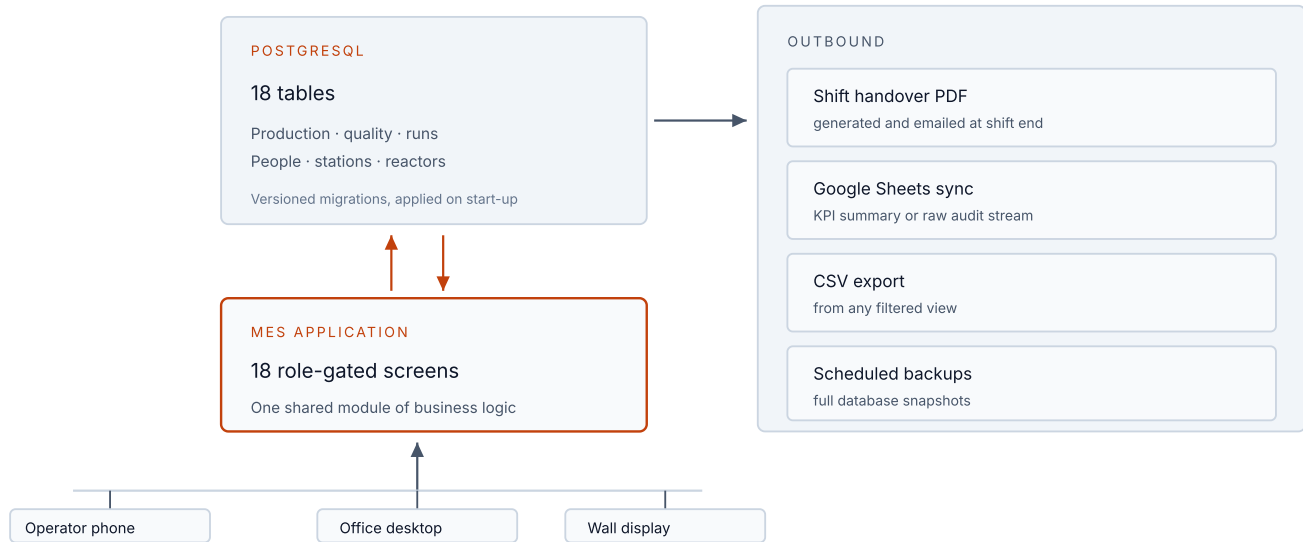
- Every unit poured, attached to an operator, a station, a resin, a lot number and a timestamp.
- A lot check against each pouring log — every container format — including the ones that failed.
- Downtime by reason and duration, and scrap split between empty and filled containers.
- Start-of-shift, changeover and spill photo audits, filed against the station.
- Pre-shift validation per operator, per shift, per pump station.
- Reactor volume, reconciled against what has been logged rather than overwritten.

### SCOPE

The system is in service for resin pouring. Pack-out logging and machine integration are built but not in use; both are described in section 14 rather than here, because neither has been committed to.

# One database, many windows

Nothing on the floor keeps its own copy of the numbers.



## Schema changes

Structural changes ship as numbered migrations that apply themselves on start-up, rather than SQL run by hand. Logs carry foreign keys to the operator and station as well as the legacy name text, so renaming someone does not split their history.

## Where the logic lives

Business logic sits in one module the whole application shares. Every screen that shows a figure reads it through the same functions, so a change to how yield is calculated applies everywhere at once.

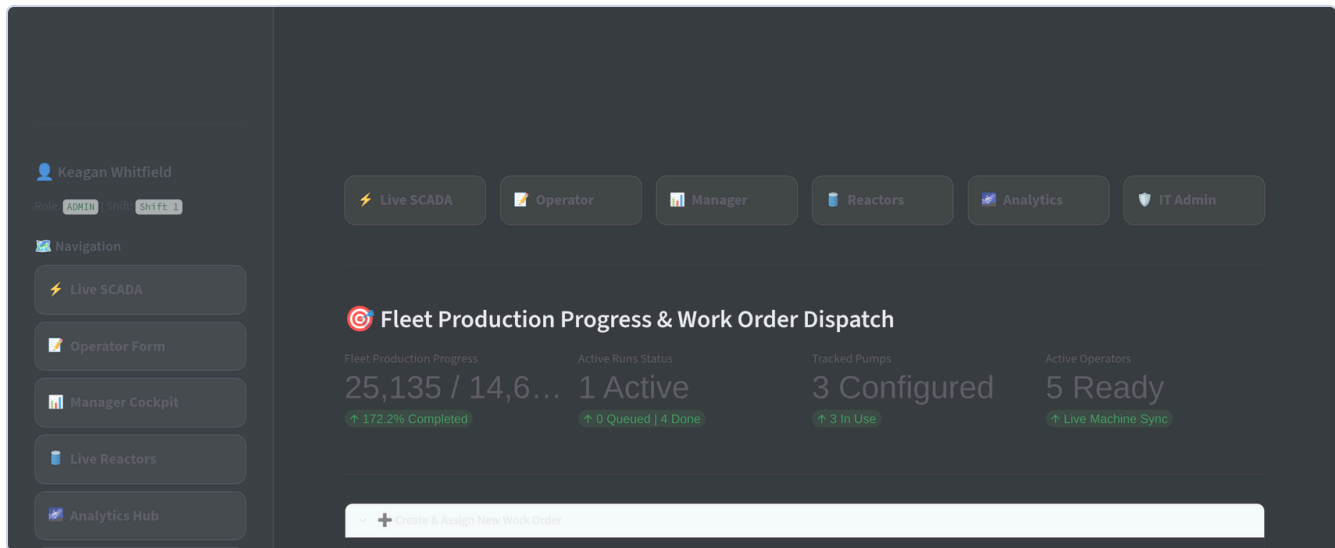
## What happens when an operator submits a log

- 1 Checked at the station**  
 The lot on the container is compared against the run before the form accepts the entry.
- 2 One transaction**  
 The log and its verification record are written together; neither can exist without the other.
- 3 The run re-counts**  
 Progress is recomputed from the logs matching its station, resin, format and lot.
- 4 Everything else follows**  
 Display, analytics, handover and exports all read those same rows.

## Work orders — optional

Off unless a plant turns it on, and the operator terminal on the next page is complete without it. This page is what becomes available if somebody wants work dispatched to stations and tracked against a target.

A manager creates a work order against a reactor, resin, container format and lot. Hourly logs submitted at that station are matched to it automatically, and the lot check gains something to compare against instead of only recording what was read.



**Work Orders & Assigned Runs.** Fleet-wide progress across active runs, with status, assigned operator, pump station and lot. Completed orders move to a separate history tab.

### DISPATCH

#### Create and assign

Reactor, size, resin, container format, target units, station and lot number, dispatched to a station rather than to one named person.

### COUNTING

#### Progress from logs

A run's count is recomputed from the production logs that match it, so deleting a bad entry re-syncs the order instead of leaving it overstated.

### SHARED STATIONS

#### Several operators, one run

Progress follows the pump station, so any number of operators can work the same run without a duplicate assignment being created for each.

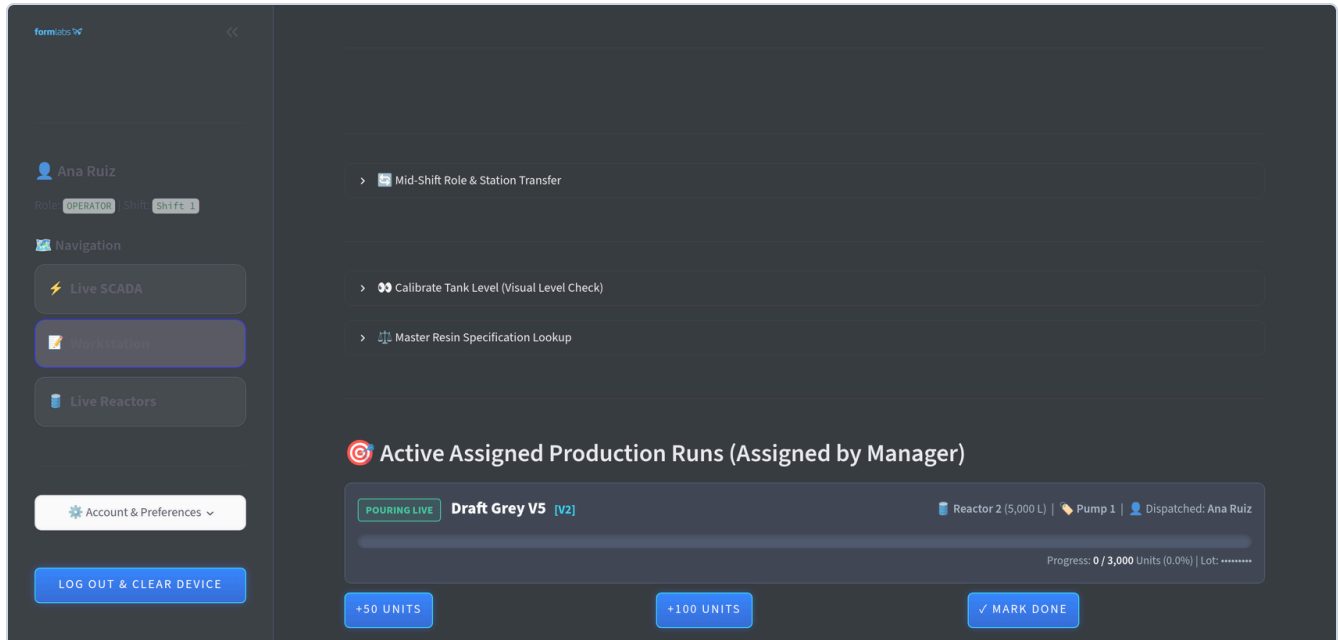
### CLOSEOUT

#### Finish on the real number

A run can be completed on a custom final total when the physical count differs, and the reactor it drew from is released.

# The operator terminal

One screen, sized for a gloved thumb on a phone: stacked inputs, large targets, no navigation to get lost in. Operators open it on their own handsets rather than at a shared terminal.



**Active runs at this station.** Reactor, container format, dispatching manager and live progress. The lot is masked for operators — see section 4.

## TAB 1

### Log hourly pouring

Station, container format, resin, verified lot, good units and both scrap categories, with the target fill weight and tolerance for the selected resin shown alongside.

## TAB 2

### Log station downtime

Reason from the plant's configurable list, duration and notes. Feeds the downtime ranking used in analytics.

## TAB 3

### Cleanliness & photo audit

Start of shift, end of shift, line transfer or spill, with a camera capture or upload, filed against the station.

## TAB 4

### Manager comms

A direct message thread with the plant lead, inside the terminal.

## ALWAYS VISIBLE

Units poured, scrap and quality yield for that operator sit at the top of the screen and update as they log. It is the same figure management sees; there is no separate scoring.

## Lot verification

Lot labels get mixed up at the pumps: a pallet of pre-labelled empties is staged that belongs to a different lot, and resin goes into it. An earlier build of this form auto-filled the run's lot and printed it on screen, which meant an hour could be logged without a cartridge ever being turned over — the check was answering its own question.

**2. Cartridge Lot Verification**

Expected lot for this run: \*\*\*\*\* hidden on purpose — read the cartridge, not the screen

Turn the cartridge over. The lot label on the bottom reads L- followed by the lot. Type it exactly as printed — spacing, case and the prefix don't matter.

L- — lot on the cartridge bottom label

L-1100-742

**The check.** The lot the run expects is masked. The operator turns the container over and types the L- code from the lot label on its bottom. Spacing, case and the prefix are normalised, so a code typed verbatim matches a run lot entered bare.

**STOP — DO NOT POUR.** This cartridge is not from the lot assigned to your run. Set it aside and get your lead.

Pulled the cartridge instead? Log the catch — it belongs in the record.

**WRONG CARTRIDGE — PULLED IT, NOTHING POURED**

Logging it anyway? Say what happened.

Choose an option

Details (required)

**When it doesn't match.** Submission is refused until the cartridge is pulled — recorded as a catch, with no production logged — or a cause, details and a photograph of the label are supplied. The log then saves flagged, carrying the lot physically in the container.

### FREQUENCY

#### Full check, then a confirm

The full check runs on the first log of a shift, on any change of run, lot, resin or station, after four hours, and on every tenth log. Otherwise a single confirmation tap.

### COST

#### Photo only on a mismatch

A clean check is one field and one tap. The photograph is required only when logging against a flag — and deliberately not required to pull a cartridge, so the safe action is never the slower one.

## 2. Jug Lot Verification

Expected lot for this run: \*\*\*\*\* hidden on purpose — read the jug, not the screen

Turn the jug over. The lot label on the bottom reads L- followed by the lot. Type it exactly as printed — spacing, case and the prefix don't matter.

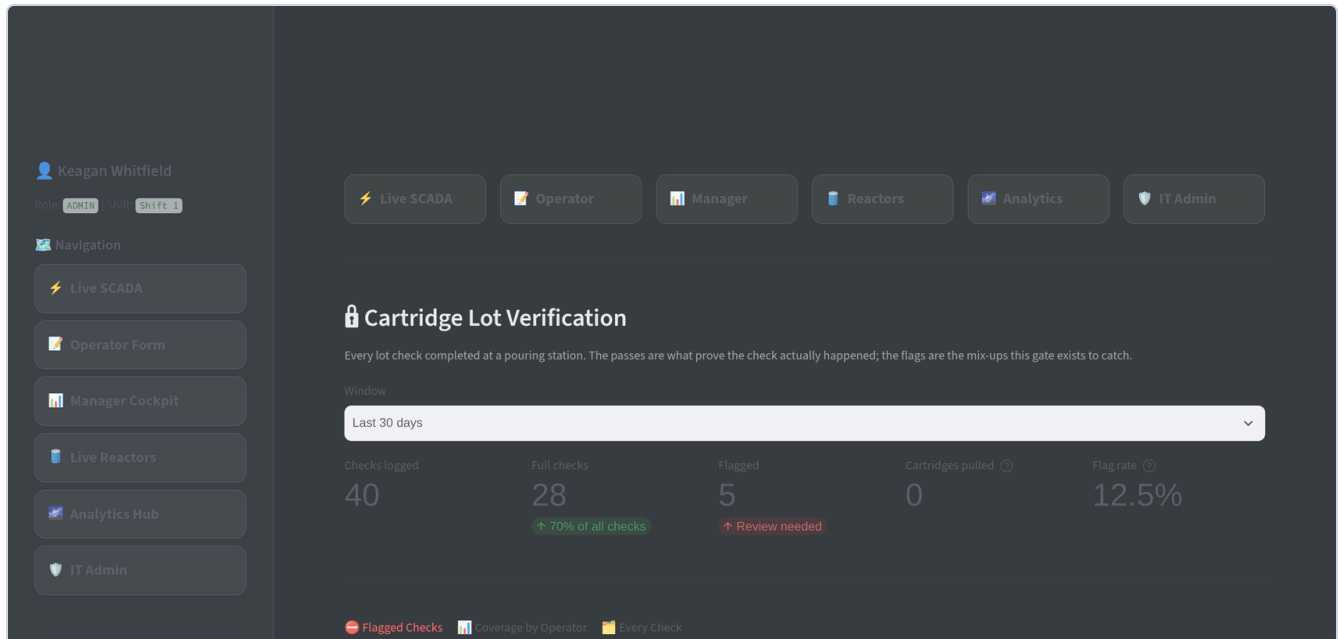
L- — lot on the jug bottom label

**The same check on a bulk jug.** RPS was the one exempt format, on the understanding that jugs carried nothing to read. They carry the same lot label a cartridge does, in the same place — so it is the same check, and only the noun changes with the format picked above.

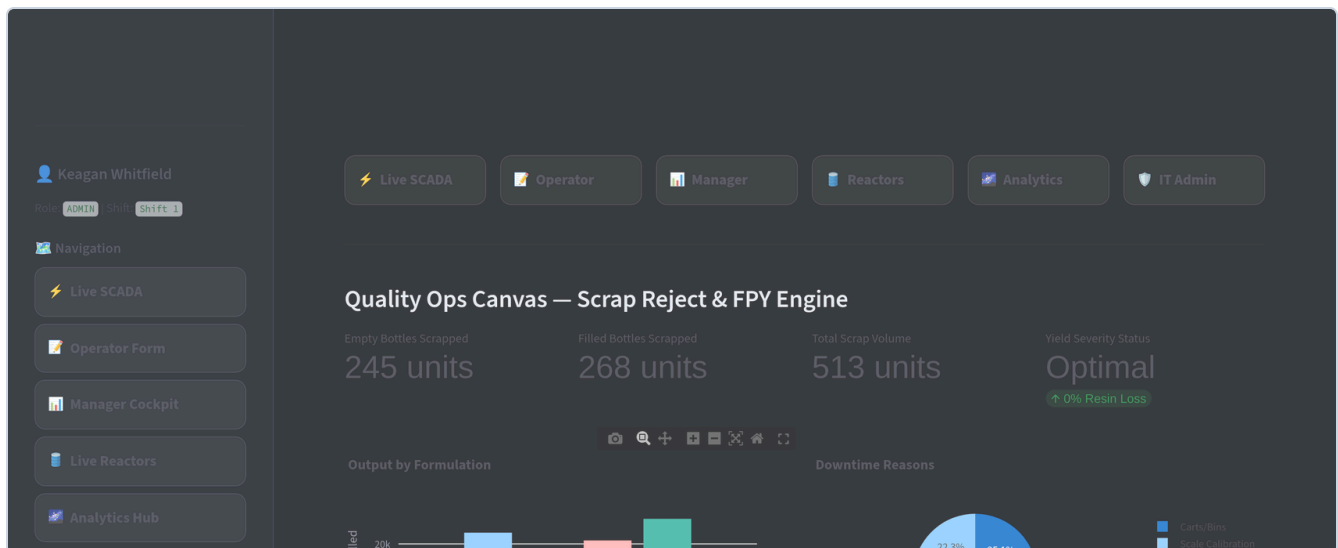
### DESIGN NOTE

A mismatch does not stop the line; it requires an explanation, because a control that halts production is one people learn to work around. Every flag and every pulled container reaches the manager's review page.

## Review and scrap intelligence



**Lot Verification review.** Every check completed, pass or fail. The passes record that the check happened; the flags are the mix-ups. Coverage per operator shows who is completing full checks and who is using the confirmation tap.



**Scrap & Yield Intelligence.** First-pass yield against the configured target, with scrap split between empty and filled containers — a handling problem and a lost-resin problem respectively — and downtime reasons ranked alongside.

### PHOTO AUDITS

#### Cleanliness gallery

Start of shift, end of shift, line transfer and spill audits with photograph, station, operator and timestamp. Spills are counted separately.

### EXPORT

#### CSV from the register

The verification register exports directly, so a quality question or customer audit can be answered without a database query.

## Fill weight and giveaway

The system has always known what every cartridge is supposed to weigh. Until now it never recorded what one actually weighed, so it could say how many units were poured but not how much resin went into them.

# 16

resin specifications, each with its own target and tolerance window

# 1

optional reading an hour — never mandatory, never pre-filled

# 4 kg

given away per thousand cartridges by a pump running 4 g heavy

### Why a passing check is not the same as a right weight

A pump set up to stay safely clear of the low limit sits *above* target on every cartridge it fills. Every one of those passes. Every one carries resin the customer did not pay for. On a 1110 g target each gram over is about 0.09% of the fill — small on one cartridge, and 4 kg per thousand at four grams heavy. Four of the sixteen specifications make it worse by being lopsided: **Grey V5** and **Clear V5** allow 10 g under target and only 5 g over, so the safe place to sit is nearer the ceiling. The giveaway does not depend on that, though — it happens on a symmetric window too. It was invisible because the target was displayed and never compared against anything.

**3. Production Output**

☒ Good Units / Containers Filled

- +

Scrap Empty
 
- +

Scrap Filled
 
- +

Check weight (g) — optional ?

✖ - +

☒ In band — 4.0 g over target.

Process Observations / Notes

e.g. Target fill weight nominal...

**One box on the hourly form.** The reading is judged against that resin's own window as it is typed. Here 1114 g on a 1110 g target is inside spec and four grams heavy — a cartridge that passes and still gives resin away.

#### OPTIONAL BY DESIGN

##### It can never block a log

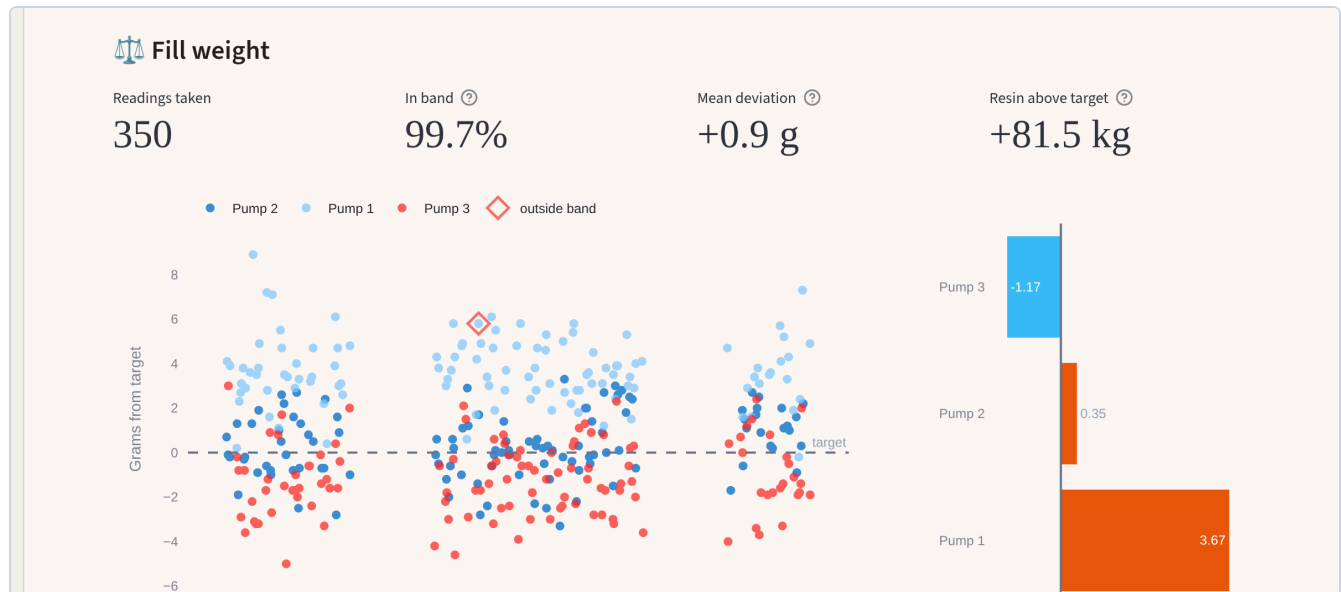
The lot gate stops the line because a wrong lot is a defect. A heavy cartridge is information. Make a weight mandatory and within a week it is the target typed from memory on every log — and a column full of **1110** is worse than an empty one, because it looks like data. Sparse and honest beats dense and invented.

#### NEVER PRE-FILLED

##### The box starts empty

For the same reason the run card had to stop printing the lot the check was hiding: a field that already contains the right-looking answer gets accepted, not measured.

# What the readings add up to



**Fill weight, across a fortnight of pouring.** Deviation from target rather than absolute grams, so resins with different targets can share an axis; zero is target and out-of-band readings carry their own marker. On the right, mean deviation per pump — one running heavy, one light, one close to nominal. That is the whole argument in one chart: every point above the dashed line passed its check.

## ONE READING, NOT THREE

### Between hours, not within a minute

Three cartridges weighed in the same minute mostly measure the scale. The variation worth catching is hour to hour, so one reading stands as a sample of its hour — and each is weighted by the units logged beside it, which is what turns “four grams heavy” into kilograms.

## JUDGED ONCE

### The verdict is stored, not recalculated

A reading is compared against the window in force when it was taken, and that verdict is kept. Editing a specification later changes what happens next; it cannot retrospectively re-judge readings somebody already took.

## ONE NUMBER THAT HAS TO STAY HONEST

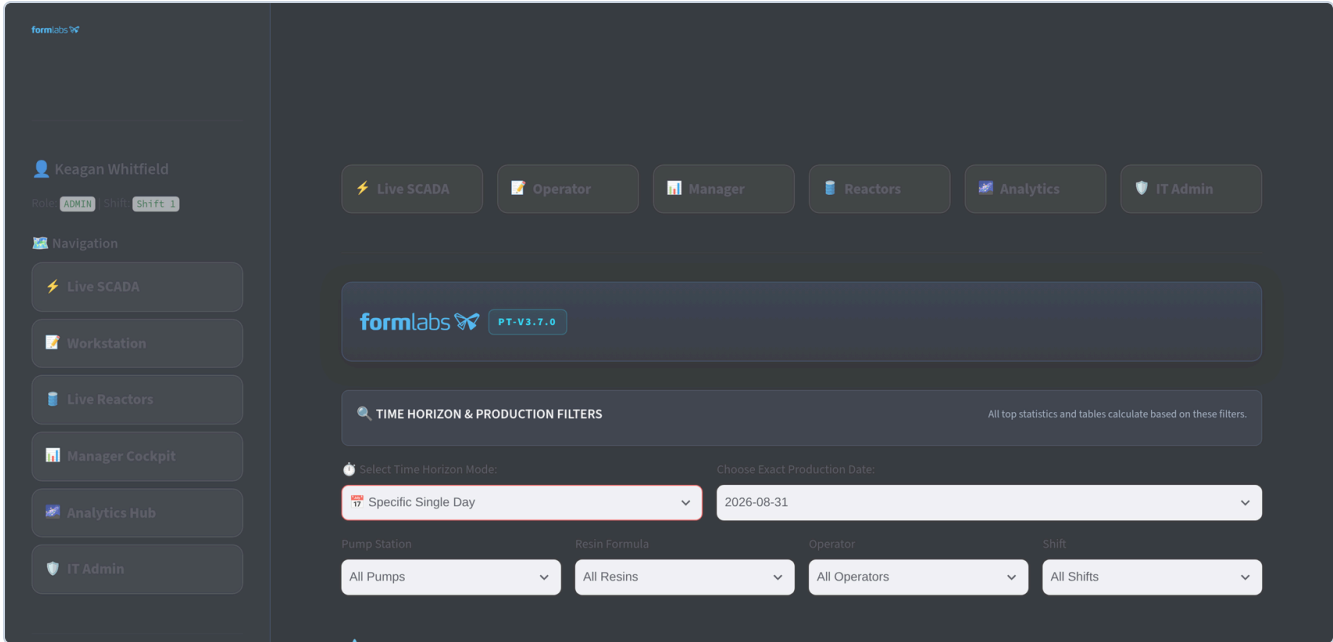
The headline share of readings inside their window never rounds a near-miss up to a clean 100%. 349 of 350 is 99.7%, and printing that as 100% directly above a chart in which the one exception is plainly visible costs the reader's trust in every other figure on the page — and that one exception is the reason to look.

## WHERE THIS LEADS

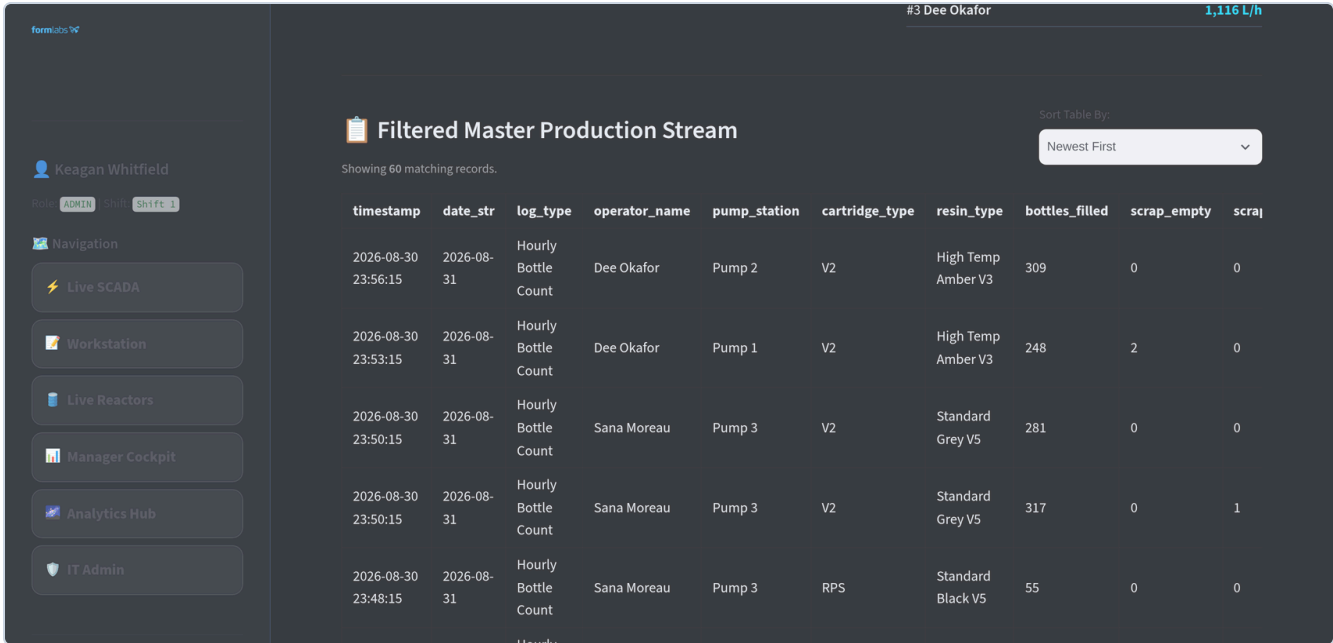
The machine-integration gateway described in section 14 already carries a serial adapter written for exactly these bench scales. If they have a serial output the weight arrives with no operator effort at all — and these manual readings are what that automated capture would have to be validated against, since checking it needs known-good numbers to check against.

# The floor as it is now

The landing screen for every role. Filters across time horizon, station, resin, operator and shift recalculate every figure on the page.

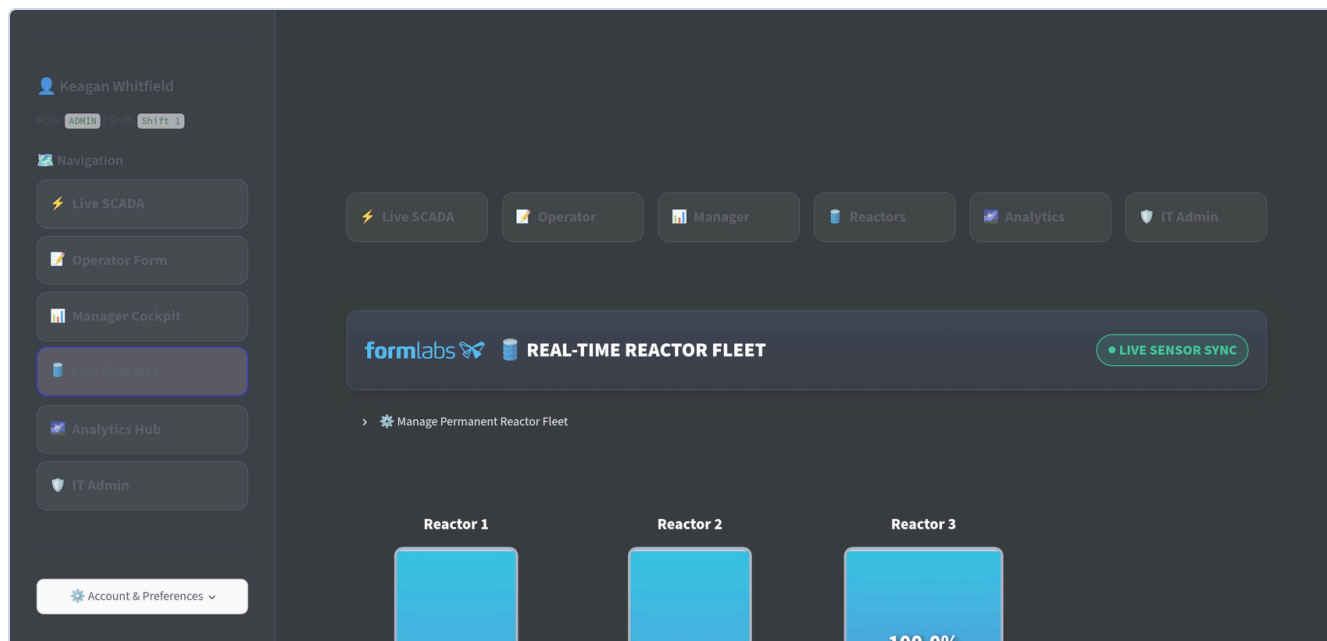


**Live SCADA.** Volume output, resin mass poured (calculated from each formulation's master fill weight), run velocity against the configured target, and pouring yield.

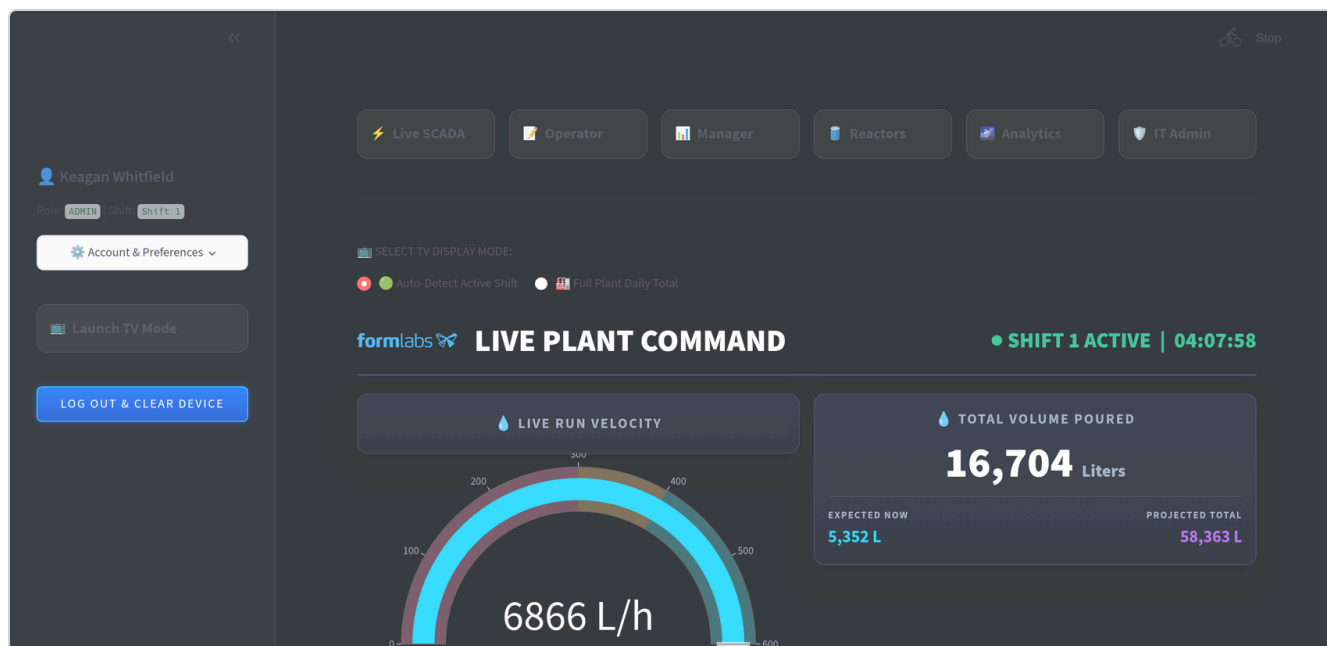


**Below the fold.** Shift trajectory against pace, a pouring leaderboard, and the filtered production stream — the individual logs behind the figures above.

## Reactors and the floor display

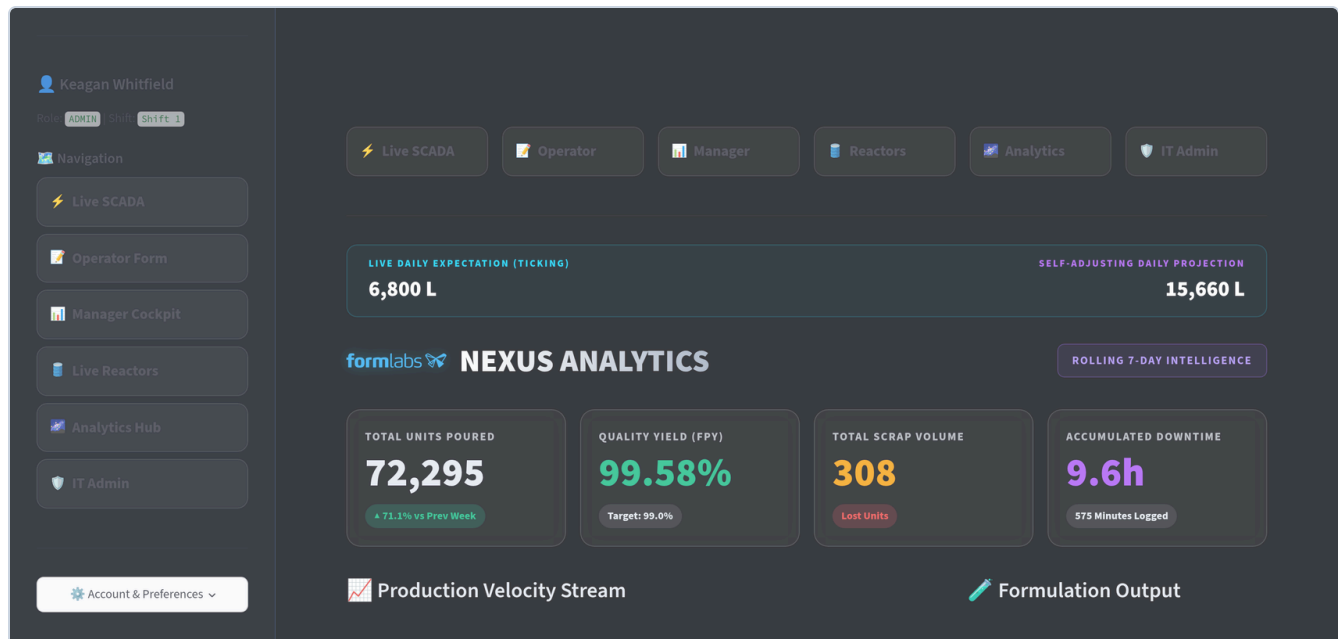


**Live Reactor Fleet.** Each vessel's current resin, assigned pump and remaining volume, with reconciliation by visual fill percentage or measured litres when the calculated figure has drifted from what is in the tank.

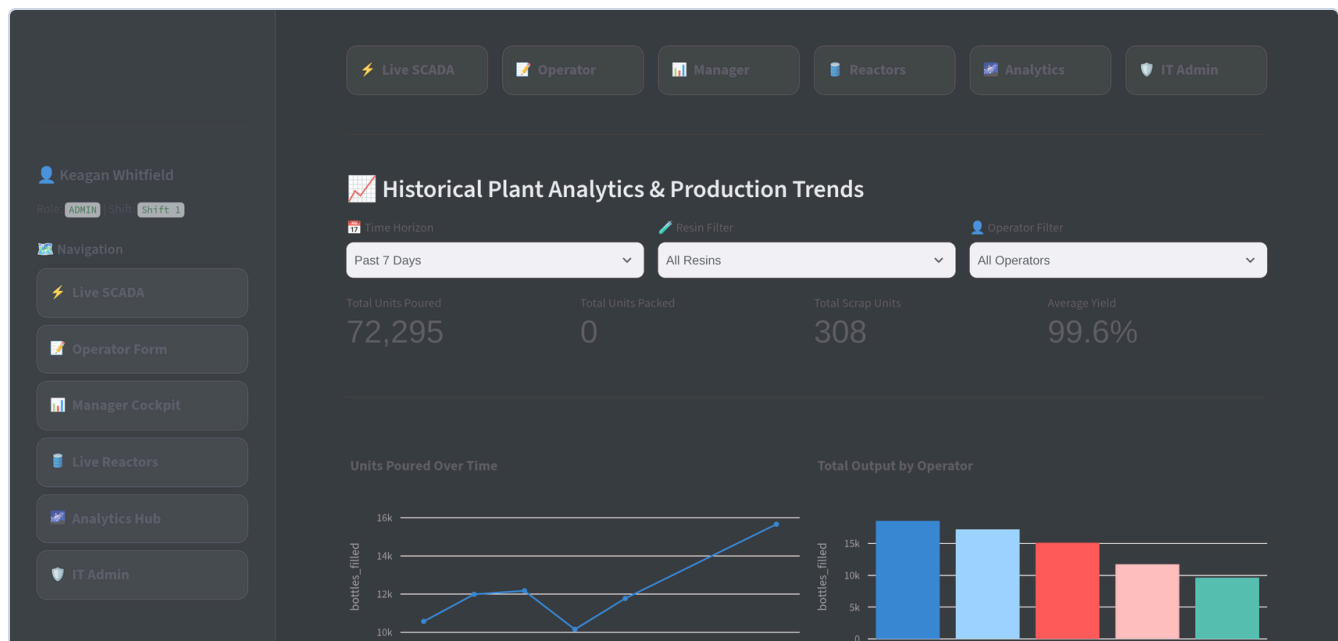


**Floor display mode.** A full-screen, self-refreshing view that auto-detects the active shift or shows the full plant day, so the team can see its own pace without opening anything.

# Trends

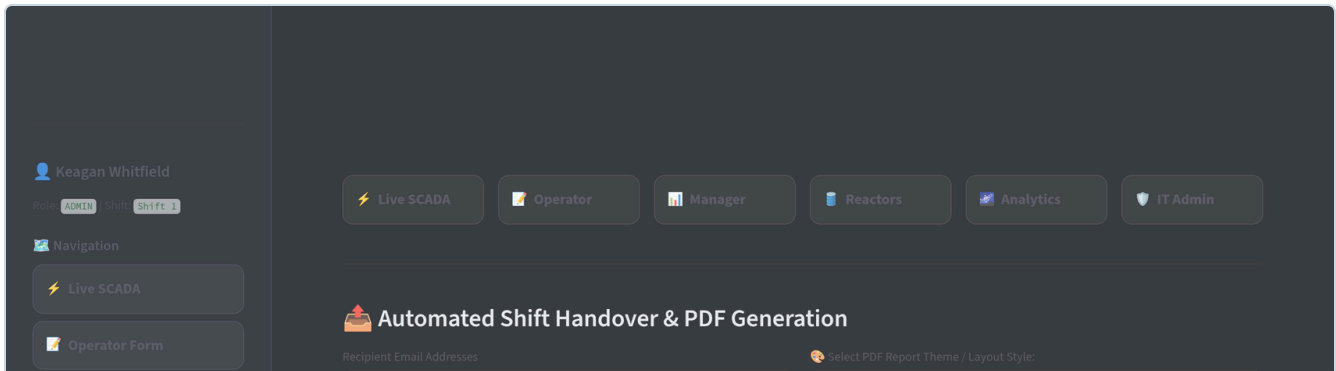


**Analytics Hub.** Output trend over the selected horizon, share of production by resin formulation, and downtime ranked by reason.



**Historical Production Trends.** The same data filtered by horizon, resin and individual operator, for month-to-date, year-to-date or any range.

# Reporting out



**Automated Shift Handover.** Output, yield, downtime and scrap for the shift, rendered as a PDF in one of two layouts and emailed to the plant's distribution list.

## CLOUD EXPORT

### Google Sheets sync

Two payload shapes — an aggregated KPI summary or the raw production audit stream — across four time horizons, pushed on demand, at shift handover, or on a scheduled webhook. Columns are selectable, and internal-only fields are stripped before anything leaves the app.

## SELF-SERVICE

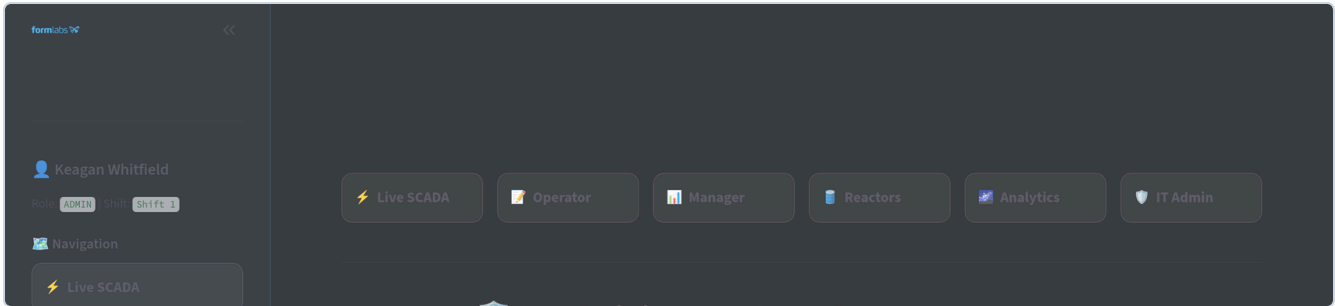
### CSV from any view

Filtered log views and the verification register export directly, so finance, quality or an audit can be answered without a developer.

## ONE SOURCE

The handover PDF, the Google Sheet, the CSV export and the floor display all render from the same production log rows. There is no second system to keep in step.

# Running the system



**IT Administration.** Roster, the issue inbox, database utilities and global plant configuration.

### PEOPLE

#### Accounts and roles

Provision staff, assign role and shift, set per-operator rate targets, reset PINs and unlock accounts. Managers can provision floor personnel without full IT access.

### PLANT

#### Global parameters

**How many shifts the plant runs** — this one runs two — with start times, lengths and breaks; rate and yield targets; handover email recipients; and the address of the pump form the operator terminal links out to.

### EQUIPMENT

#### Stations, reactors, reasons

Pump stations, the reactor fleet and capacities, and the downtime reason list, so the vocabulary stays the plant's own.

### RECOVERY

#### Backup and restore

Full database snapshots on demand or on schedule, and a restore path from any snapshot.

Quick Search by SKU, Resin Name, or Code

Showing: ALL (16 Resins)

| ID | FORMAT | SKU           | CODE     | RESIN NAME      | TARGET (G) | MIN (G) | MAX (G) | TARGET (KG) | MULTIPLIER | LIFETIME |
|----|--------|---------------|----------|-----------------|------------|---------|---------|-------------|------------|----------|
| 8  | RPS    | RS-R5-GPBK-05 | FLGPBK05 | Black V5        | 5000.0     | 4950.0  | 5050.0  | 5.0         | 1.0        | 24       |
| 6  | V1     | RS-C1-DRV2-01 | FLDRV201 | Draft V2        | 1120.0     | 1110.0  | 1130.0  | 1.12        | 1.0        | 24       |
| 11 | V1     | RS-C1-FL80-01 | FLFL8001 | Flexible 80A V1 | 1070.0     | 1060.0  | 1080.0  | 1.07        | 1.0        | 24       |
| 7  | V1     | RS-C1-TO15-01 | FLT01501 | Tough 1500 V1   | 1070.0     | 1060.0  | 1080.0  | 1.07        | 1.0        | 24       |
| 16 | V2     | RS-C2-BIOX-01 | FLBIOX01 | Bioresin XZ9    | 1050.0     | 1040.0  | 1060.0  | 1.05        | 1.0        | 24       |
| 15 | V2     | RS-C2-BLK6-01 | FLBLK601 | Black V6        | 1110.0     | 1100.0  | 1120.0  | 1.11        | 1.0        | 24       |

**Master Resin Specifications.** Target fill weight, tolerance band and shelf life — what every check weight is judged against, and what turns a bottle count into resin mass. Each formulation carries its colour from the plant's own master sheet.

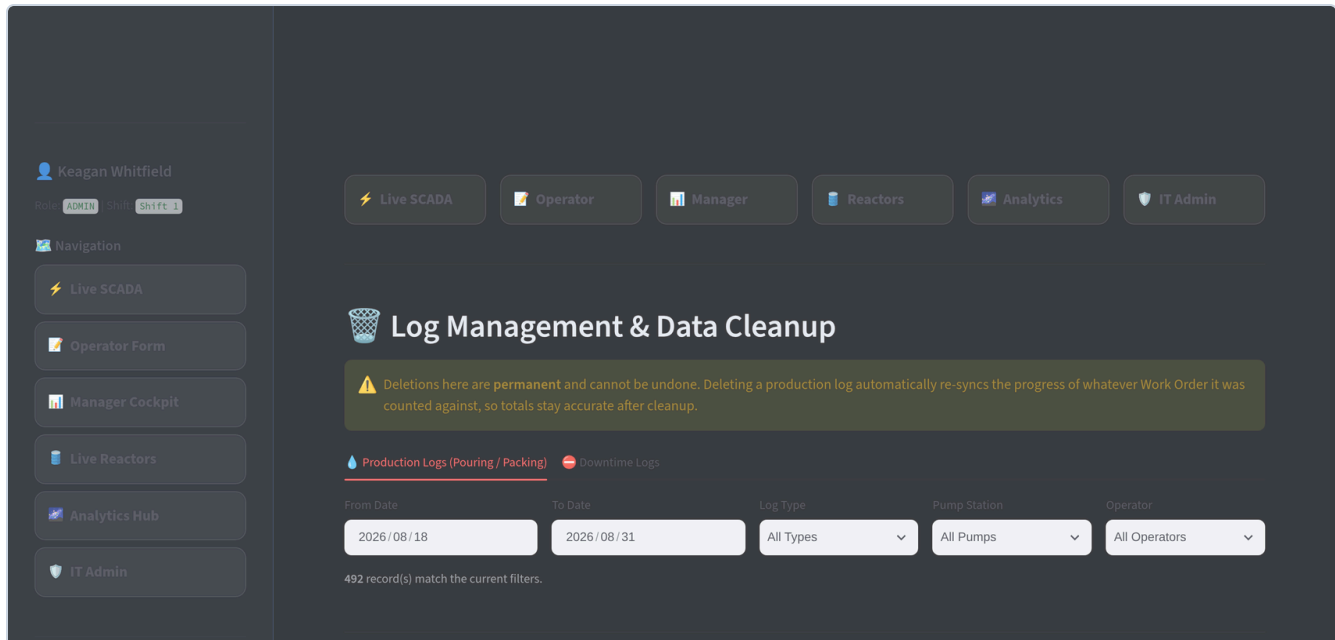
# Access control and data integrity

## Accounts

- **PINs are hashed.** Salted with bcrypt; the database holds no readable PIN, including for administrators.
- **Lockout.** Five failed attempts locks the account for fifteen minutes, with an instant IT unlock.
- **Sessions are tokens.** A signed token with a fixed lifetime backs "remember this device", revocable server-side.
- **Every page checks the role.** Enforced on the page itself, so typing a URL for a screen your role lacks returns access denied.
- **Output is escaped.** Values typed on the floor are escaped before rendering, so a note cannot inject markup or break its own card.

## The record

- **History survives a rename.** Logs carry foreign keys to the person and station as well as the name text.
- **Deletions re-sync.** Removing a log recalculates whatever work order it counted toward.
- **Bulk deletion is deliberately hard.** Disabled until the word DELETE is typed into a confirmation box.
- **Corrections are visible.** A plant-level correction is written as a labelled system adjustment, not by editing an operator's entries.
- **Credentials are kept out of the application itself.** Connection strings and passwords live in a configuration file on the machine, not written into the program.



**Log Management & Cleanup.** Filter production and downtime logs by date, station, resin or operator, review, and delete with the work-order re-sync handled automatically.

# Roles

Everyone signs into the same application and sees a different one. Access is checked on every page, so the difference is enforced rather than cosmetic.

| CAPABILITY                                    | OPERATOR | MANAGER     | ADMIN     |
|-----------------------------------------------|----------|-------------|-----------|
| Live SCADA dashboard                          | Yes      | Yes         | Yes       |
| Startup checklist & terminal unlock           | Required | n/a         | n/a       |
| Log hourly pouring & lot verification         | Yes      | Test mode   | Test mode |
| Log downtime & cleanliness audits             | Yes      | Yes         | Yes       |
| Message the plant lead                        | Yes      | Both ways   | Both ways |
| Live reactor fleet & reconciliation           | Yes      | Yes         | Yes       |
| See the lot a run expects                     | Masked   | Yes         | Yes       |
| Create & dispatch work orders                 | —        | Yes         | Yes       |
| Analytics hub & historical trends             | —        | Yes         | Yes       |
| Lot verification review & scrap intelligence  | —        | Yes         | Yes       |
| Shift handover PDF & Google sync              | —        | Yes         | Yes       |
| Floor roster & PIN resets                     | —        | Floor staff | All users |
| Master resin specifications                   | —        | Yes         | Yes       |
| Log management & bulk cleanup                 | —        | Yes         | Yes       |
| Plant configuration, shift count & parameters | —        | —           | Yes       |
| Record a check weight against an hourly log   | Optional | Optional    | Optional  |
| Undo your own last log, within two minutes    | Own only | Own only    | Own only  |
| Database backup & restore                     | —        | —           | Yes       |

*Test mode* lets a manager or administrator log on behalf of an operator for training and troubleshooting. The entry is attributed to the operator selected and is visible in the log.

# Your shift, start to finish

|                                                  |                                          |                                 |
|--------------------------------------------------|------------------------------------------|---------------------------------|
| <b>Operator</b><br>Pours resin at a pump station | <b>Manager</b><br>Dispatches and reviews | <b>Admin</b><br>Runs the system |
|--------------------------------------------------|------------------------------------------|---------------------------------|

## 1 Sign in on your phone

Operators work from their own handsets rather than a shared station terminal. Open the plant address in your browser and sign in with your operator ID and PIN, then add it to your home screen so it is one tap next hour. Tick *Remember this device* — on a phone only you carry that is the right answer, and it means you are not typing a PIN every time you log an hour. Five wrong PINs locks the account for fifteen minutes — ask your lead for a reset rather than guessing.

## 2 Clear the startup checklist for your pump

The terminal stays locked until you do. Pick the pump you are starting at, submit the start-of-shift cleanliness photo for that station, then confirm the QC checksheet and that your empties and receiving carts are staged. **Moved to a different pump later? The checklist comes back for that pump** — it certifies the station you are standing at.

**Terminal locked.** Choose your pump here; the answer carries through to your logging tab, so you pick your station once.

## 3 Check the run you have been assigned

Runs assigned to your station appear at the top with reactor, container format and live progress. If nothing is listed, tell your lead before pouring — your units will still be recorded, but they will not count toward a work order.

### IF YOU ARE LOCKED OUT

#### Five attempts, fifteen minutes

The lock releases on its own, but a manager can reset your PIN from the floor roster immediately.

### YOUR OWN PHONE

#### Nobody else logs under your name

Because it is your handset you can stay signed in between shifts — there is no next operator picking it up. What matters instead is that every log is attached to whoever is signed in, so do not hand your phone to someone else to log their hour. If you lose it, tell your lead and have your PIN reset.

# Logging an hour of pouring

## 4 Set your station and material

Pump station, container format (V2, V1, RPS bulk jug or Pigment) and resin formulation. The target fill weight and tolerance for that resin appear underneath — check it against your scale.

## 5 Verify the lot

Turn the container over and read the **L-** code from the lot label on its bottom — the same label on a cartridge and on a jug. The lot the run expects is hidden on purpose — **read the container, not the screen**. Spacing, case and the prefix do not matter.

**2. Cartridge Lot Verification**

Expected lot for this run: L-2502T0310 hidden on purpose — read the cartridge, not the screen

Turn the cartridge over. The lot label on the bottom reads **L-** followed by the lot. Type it exactly as printed — spacing, case and the prefix don't matter.

L- — lot on the cartridge bottom label

☒ Lot matches this run.

**A clean check.** Green means the cartridge belongs to this run. On the next log at the same station with nothing changed, this becomes a single confirmation tap — returning to the full check on any change of run, lot, resin or station, after four hours, and on every tenth log.

## 6 If it does not match, stop

A red **STOP — DO NOT POUR** means the cartridge is not from your run's lot. Set it aside, tell your lead, and press *Wrong cartridge — pulled it, nothing poured*. That records the catch. If your lead decides the pour goes ahead, choose the reason, add the detail and photograph the lot label — the log saves with the flag attached.

## 7 Enter your counts and submit

Good units filled, plus scrap empty and scrap filled kept separate — they mean different things to quality. Add anything worth noting, then submit. Your totals and the run's progress both move immediately, and the screen says plainly whether the entry reached the database — if it did not, nothing was recorded, so submit again rather than wondering.

## 8 Weigh one if you have the scale to hand

**Optional, every time.** One cartridge, one number, in the *Check weight* box; it says straight away whether that cartridge is inside its window and how far off target. Leaving it blank never blocks anything — a blank is honest, a guess is worse than nothing.

## 9 Typed the wrong number? Take it back

Two minutes to undo your own most recent log — a 2500 that should have been 250 no longer needs a manager. Your latest entry only, and never a flagged one.

# The rest of your terminal

## WHEN THE LINE STOPS

### Log the downtime

Reason, minutes, and anything useful. Log it as it happens rather than reconstructing it later: downtime you do not record shows up as poor yield instead of a machine problem, and the ranking management uses to prioritise fixes is built from these entries.

## CHANGEOVERS AND SPILLS

### Photo audits

Start of shift, end of shift, moving a line onto a different resin, or a spill. These are the record that a changeover was done cleanly.

## QUESTIONS

### Manager comms

A direct thread with the plant lead inside the terminal, so the answer stays attached to your account rather than living in a text message.

## YOUR NUMBERS

### Today's performance

Units poured, scrap and quality yield at the top of your screen, updating as you log.

## THE PUMP FORM

### A button, not a walk to the sticker

If a pump form has been configured, a button at the top of the terminal opens it in a second tab. The MES stays open behind it, so you finish the form, close it, and are back mid-log with nothing retyped — which scanning the QR sticker cannot do, because that navigates away from the MES.

## WORTH KNOWING

**Your logs are attached to you**, and nobody can quietly overwrite one.

**Every container format is checked, RPS jugs included.** Jugs carry the same lot label in the same place, so it is the same job — turn it over, read the code. The screen says jug or cartridge to match the container format you picked.

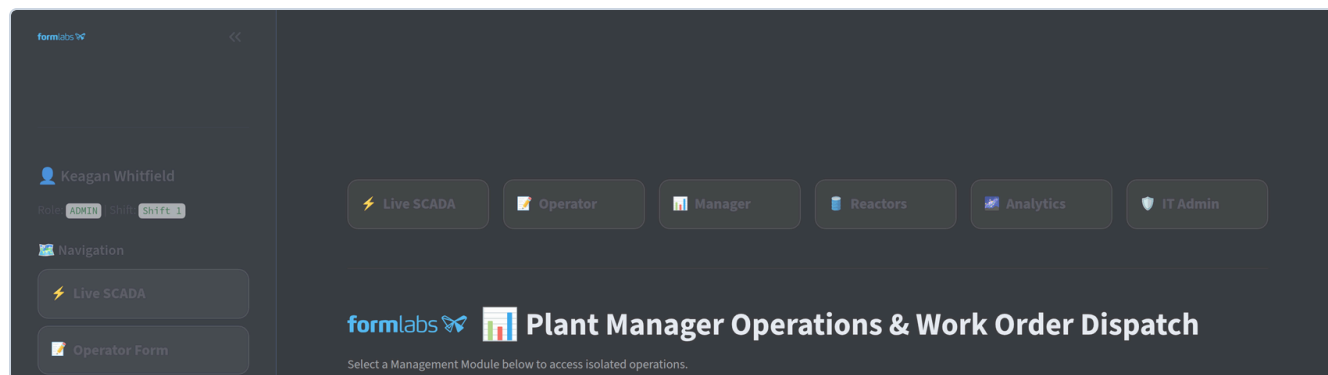
**Nothing you submit is silently discarded.** If a log cannot be matched to an active run you are told on screen, rather than finding out later that a progress bar never moved.

## THE ONE RULE

Log it when it happens. An hour reconstructed at the end of a shift is a guess, and every pace figure, yield number and downtime decision downstream is built on what you enter.

# Dispatching the day

|                                                  |                                          |                                 |
|--------------------------------------------------|------------------------------------------|---------------------------------|
| <b>Operator</b><br>Pours resin at a pump station | <b>Manager</b><br>Dispatches and reviews | <b>Admin</b><br>Runs the system |
|--------------------------------------------------|------------------------------------------|---------------------------------|



**Manager Cockpit.** Each management module on its own screen rather than one page trying to do everything.

1

## Create the work orders

Reactor and size, resin, container format, target units, pump station and lot number. Assign an operator for accountability, but note progress follows the *station* — anyone working that pump contributes to the same run.

2

## Enter the lot exactly as printed

The most important field on the form: it is what the operator's cartridge check is compared against. Enter the number as printed after **L -** on the base. Spacing, case and the prefix are normalised, but a wrong number cannot be.

3

## Watch the fleet, not the logs

The work order page rolls up fleet progress, active and queued counts, tracked pumps and operators ready. Drill into a run only when its progress stalls.

### IF A PROGRESS BAR IS NOT MOVING

Almost always one of three things: the operator's station does not match the run's, the resin or container format differs, or the lot on the run does not match what is staged. The verification review shows which.

## Reading the floor, closing the day

### DAILY

#### Flagged lot checks

Mismatches poured anyway and cartridges pulled, each with the photograph, both lot numbers and the operator's reason. A mismatch that is really a formatting difference means the run's lot needs correcting, not the operator.

### DAILY

#### Scrap and downtime

Yield against target, the empty-versus-filled split, and downtime by reason. Empty scrap is a handling problem; filled scrap is lost resin and considerably more expensive.

### WEEKLY

#### Verification coverage

Full checks versus confirmation taps per operator. A station showing almost no full checks is one taking the fast path every hour.

### WEEKLY

#### Trends

Output trend, share by resin, and per-operator history. The historical page covers anything beyond the current fortnight.

4

#### Close out finished runs

Complete a run on its real final count when the physical number differs. The correction is booked as a labelled system adjustment and the reactor is released.

5

#### Send the handover

Generate the PDF and dispatch it to the recipient list. Use the print-friendly layout for anything read on paper or forwarded outside the plant.

6

#### Answer the floor

The communications page holds every operator's thread. A question left unanswered there gets solved badly on the floor instead.

# Keeping it running

## Operator

Pours resin at a pump station

## Manager

Dispatches and reviews

## Admin

Runs the system

### ON JOINING

#### Provision the account

Create the user with the correct role and shift before their first shift. Role decides which screens exist for them; shift decides which checklist and pace target apply.

### ON LEAVING

#### Remove access, keep history

Deleting a user removes access. Production history stays attached through the foreign key on every log.

### WEEKLY

#### Verify a backup

Take a snapshot and confirm it lands. A backup that has never been restored is a hypothesis; the restore path is on the same page.

### PER RELEASE

#### Let migrations run

Schema changes apply on start-up. After deploying, start the app once and confirm it comes up before the shift arrives.

### SEASONALLY

#### Review plant parameters

Shift times, lengths, break minutes, rate and yield targets. Every pace indicator and yield judgement is measured against these.

### MONTHLY

#### Clear the suggestions inbox

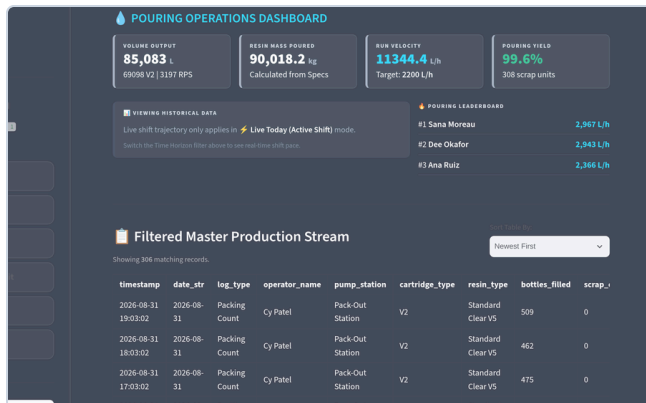
Staff file requests, bugs and floor issues from inside the app. An inbox nobody empties is one nobody writes to.

### NOT THIS

Do not edit production rows directly in the database to fix a number. Use log management to remove the bad entry — it re-syncs the affected work order — or a reconciliation to book a variance. Both leave a visible record; a direct edit leaves the totals right and the story wrong.

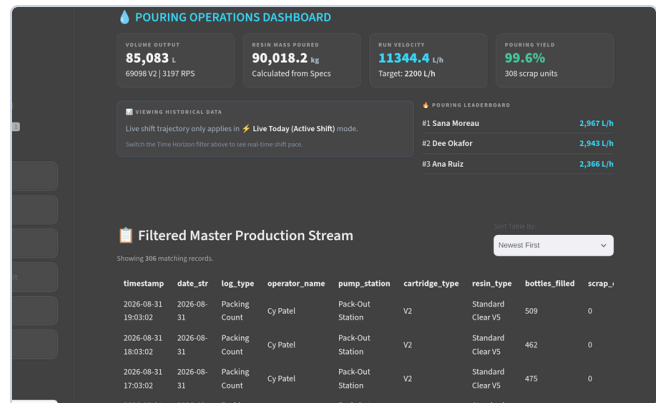
# Thirty-four themes

Selectable per user and remembered across sessions. This is less cosmetic than it sounds: a phone held under bright shop lighting, an office monitor and a wall display twenty feet from the viewer are three different readability problems. **Formlabs Forge** is the house theme and the one this document is set in.



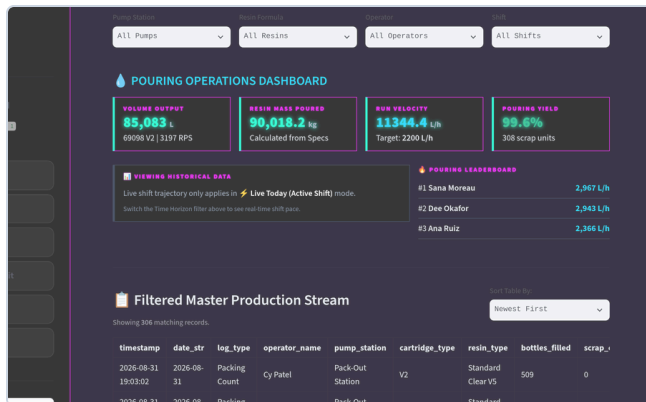
Formlabs Forge

#0F172A



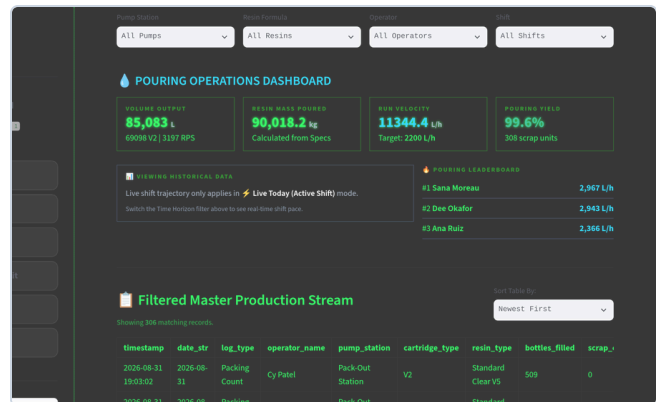
Formlabs Onyx

#0B0B0C



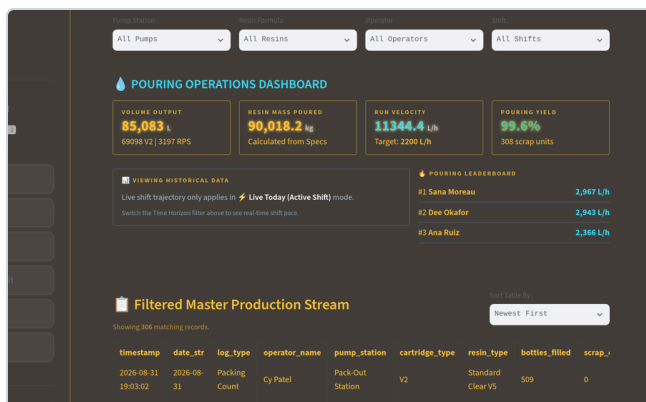
Neon Cyberpunk

#090117



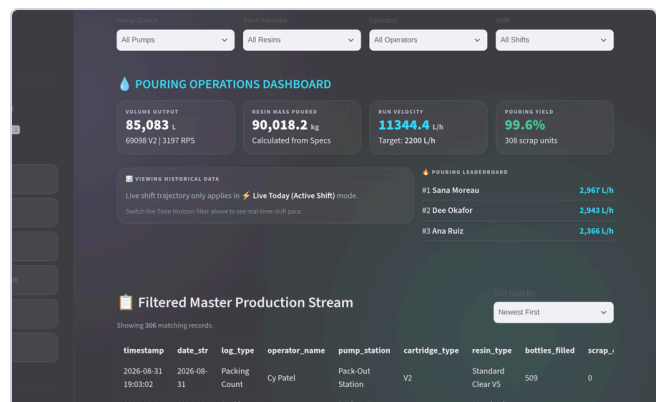
The Matrix

#000000



Amber CRT

#0D0800



Glass Aurora

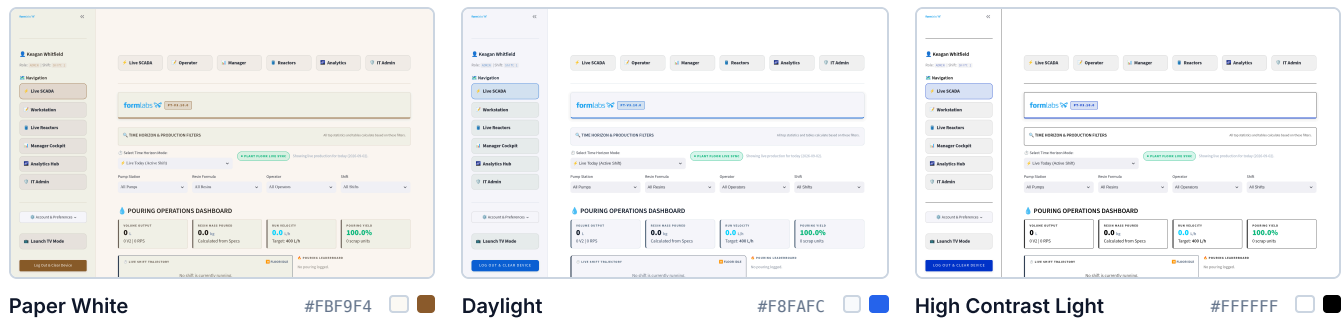
gradient



Six of the twenty-four dark themes, all rendering the same Live SCADA screen. The rest include Deep Space, Titanium Alloy, Carbon Fiber, Resin Flow, Powder Bed, Ocean Depths, Dracula and Obsidian Chrome. Light themes overleaf.

## Reading the screen on the floor

Six of the thirty-four themes are light, and the rest of this page is about the other half of the problem: a phone held in a gloved hand, at arm's length, in a room lit for pouring resin.



Three of the six light themes, all rendering the same Live SCADA screen as the dark ones on the previous page. The others are **Forge Light**, **Blueprint** and **Clean Room**. Four dark themes were added alongside them: **Nordic Frost**, **Solarized Deep**, **Workshop Retro** and **High Contrast Dark**.

### CONTRAST

#### Checked, not assumed

Every theme is measured against the accessibility floor — 4.5:1 for body text. The check found two real failures in themes already in use: card labels in **Default Dark**, the one most people are looking at, and in **Dracula**. Both fixed. All thirty-four now pass, and a Theme Gallery screen re-runs the audit on demand, so it is something you can see rather than something that was checked once.

### GLOVES

#### Glove mode

Nitrile still registers on a touchscreen, but precision drops, and the pouring form asks for number steppers and a lot field. One toggle pushes every target past the accessibility touch size and scales the type. It is remembered against the *terminal*, not the account — being gloved is a property of where you are standing, not of who you are.

### NIGHTS

#### Automatic dimming

Derived from the plant's own configured shift times rather than a fixed hour, because "night" here means "not the day shift". A gentle brightness and warmth reduction rather than a different set of colours, so it works with all thirty-four themes. Photographs are exempt, so a lot label under review keeps its true colour.

### MID-POUR

#### Focus mode

During a pour the operator is at the pump, not at the screen. One toggle strips the page to the four things that matter mid-run — resin, lot, count so far, how many left — at a size readable from across the station. A toggle rather than a separate screen, because the moment they need it is the moment they need to log.

## Resin colour, and why colour alone was not enough

Wherever a resin is named it carries the colour from the plant's own master sheet, so screen and sheet read as one document; a new formulation takes its family's colour from its name. But five pairs in that palette are the same colour to the eye, and a Clear/Rigid mix-up is exactly what the lot gate exists to catch — so each chip also carries a **texture** from the resin family: Clear V4 and Clear V4.1 match, Clear and Rigid cannot. Texture survives greyscale printing and colour blindness.

## How it is tested

Before a release the system is exercised end to end against a throwaway database built from nothing each time. It refuses to run against the production connection.

### 839

assertions across eight suites — workflow, interface, colour, fill weight, shifts and display, links, boot paths

### 23

checks driving a real browser through an operator's shift, on a desktop and a phone

### 18

screens rendered and checked for errors on every pass

### What the simulation covers

All eight migrations are applied to an empty database and the result compared table by table against the application's own model definitions. A manager then creates three work orders; operators clear per-station checklists and pour — clean logs, a fast-path log, a mismatched cartridge, a cartridge pulled before pouring, a log with no run behind it, and a gated RPS jug — alongside downtime. Every figure the interface displays is then recomputed from the raw rows and compared against hand-figured expectations: poured, scrap, yield, per-operator totals, run progress, flag rate and variance.

### What that has caught

- **The run card was printing the lot the check was hiding.** An operator could have read the expected lot off the screen instead of the cartridge, which would have made the whole check decorative. Found because the interface test asserts the lot appears nowhere on the page.
- **Plant settings could never be saved.** The Admin Panel's save button passed a dictionary to a function expecting fifteen positional arguments, so shift times, rate targets and yield targets had never been changeable from the interface.
- **A content-delivery fetch could take the terminal down.** An unguarded network call for a decorative animation ran on every interaction; a floor PC that lost its internet would have hung the terminal before a control rendered.
- **Eight pages failed on a theme change,** and log cleanup crashed whenever a filter matched fewer than ten records — both long-standing.
- **A setting that saved and was never read back.** The number of shifts had a column, a migration, an admin field and a working save path — and the function every screen reads settings through never returned it, so all of them silently used the default. It happened to be the right answer for this plant, which is why nobody noticed.
- **Every light theme rendered its stylesheet as visible text.** Found in a browser after the tests had passed: an indented style block appended after other content is treated as a code listing and displayed rather than applied. The twenty-four older themes were never bitten because theirs comes first.

#### CLOSED SINCE THE LAST ISSUE

The page sweep renders every screen but did not submit forms, which is how the settings save stayed broken. The interface suite now drives the real pouring form through submission — including the two rules that matter most about the check weight, that it never blocks a log and is never pre-filled — and a screenshot comparison catches a card that lost its border or a chart that came back empty, neither of which raises an error.

## Experiments, not commitments

Two things exist in the codebase that are **not in use and not decided on**. They are described here so the capability is known, not because there is a plan attached to either. Both were built to find out whether the approach works.

### Pack-out logging

A packing role, a pack-out logging tab, and skid conversion using each resin's units-per-skid figure. Poured and packed totals are counted separately, and the gap between them reports as floor work-in-progress. A reconciliation flow books the variance between what was poured for a lot and the final packed count as a single labelled system adjustment.

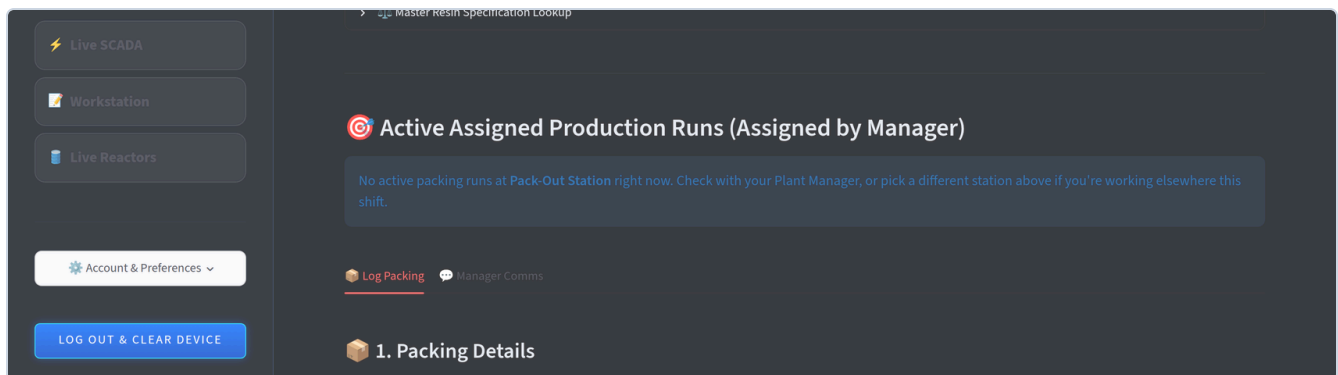
It is switched off. Turning it on is a single plant setting, but nobody has decided whether pack-out should be logged this way, and the workflow has not been tested against how the line actually runs.

### Machine integration

A protocol-agnostic gateway for reading floor equipment directly, so counts and weights would not have to be typed. Each machine is a row in a registry and the polling logic sits behind one shared interface, with adapters written for Modbus TCP, Modbus RTU, OPC-UA, MQTT, Serial ASCII and HTTP. Readings would write through the same logging path the operator screen uses, so analytics and run tracking would need no changes.

#### None of it has been tested against real equipment.

The adapters were written to cover whatever the scales or pump controllers turn out to speak; which of them is correct is unknown until someone connects one.



**Pack-out logging, currently disabled.** Container format, resin, lot and units packed, with the skid equivalent calculated as the count is typed.

#### READ THIS AS

Neither of these is a roadmap item or a commitment of dates. They are experiments that happen to be finished enough to demonstrate, and either could be abandoned without affecting anything in service.